

ANSH SINGH

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SUMMARY

Full-stack developer with strong expertise in scalable backend architectures, modern frontend frameworks, and system design. Experienced in building production-grade MERN applications, real-time systems using WebRTC and Socket.IO, and optimizing database and API performance for high reliability.

EDUCATION

Vellore Institute of Technology, Chennai (VIT Chennai)

Bachelor of Technology in Computer Science and Engineering |

2023–2027

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Java, JavaScript (ES6+), TypeScript, SQL, Bash
- **Frontend:** React.js, Next.js, Redux Toolkit, Tailwind CSS, Material UI, React Native, HTML5/CSS3
- **Backend:** Node.js, Express.js, Flask, Django, RESTful APIs, Microservices, System Design (LLD/HLD)
- **Databases:** MongoDB, MySQL, PostgreSQL
- **Data Science / ML:** NumPy, Pandas, Matplotlib, Scikit-learn, Machine Learning, Data Analysis
- **Tools & DevOps:** Git/GitHub, Docker, Postman, Swagger, CI/CD, Linux
- **Cloud & Realtime:** AWS (S3), Cloudinary, JWT, WebRTC, Socket.IO, Hugging Face
- **Methodologies:** Agile, Scrum

EXPERIENCE

AICTE

Feb 2026 – Mar 2026

Power BI Intern

- Planned and developed **interactive dashboards** processing large datasets, reducing manual reporting effort by ~40% across 3+ business verticals.
- Built **data transformation pipelines** using Power Query, improving data consistency and reducing preprocessing errors by ~30%.
- Utilized **DAX functions** to create optimized calculated fields, improving report refresh performance by ~25%.
- Architected a normalized data layer that streamlined data integration across multiple sources, improving analytical accuracy and report maintainability.
- Reduced dashboard latency by restructuring data retrieval workflows and eliminating unnecessary query executions across reporting pipelines.

PROJECTS

StudyNotion | EdTech SaaS Platform | **MERN, Redux, Socket.IO, Razorpay**

[GitHub Repository]

- ❖ Built a **full-stack Learning Management System** supporting role-based dashboards, video processing, and course monetization.
- ❖ Incorporated **Razorpay Payment Gateway** with custom webhook logic for automated enrollment and invoice generation.
- ❖ Visualized business metrics using **Recharts**, enabling instructors to monitor student retention and revenue growth in real-time.
- ❖ Leveraged **Cloudinary CDN** for global asset delivery, resulting in **40% faster page load times**.
- ❖ Established **JWT-based authentication** with secure HttpOnly cookies and session persistence, enhancing platform security.

MediConnect | AI Telemedicine Solution | **React, Node.js, WebRTC, Socket.IO, Hugging Face** [GitHub Repository]

- ❖ Developed a **real-time Consultation Engine** via **WebRTC**, facilitating encrypted peer-to-peer video streaming and chat.
- ❖ Integrated **Hugging Face Transformers** for an automated medical triage assistant, achieving **85% accuracy** in symptom analysis.
- ❖ Implemented high-security standards for patient data using **AES-256 encryption** and JWT-based session management.
- ❖ Designed an intuitive dashboard for healthcare providers to track appointments and digital health records (EHR).
- ❖ Orchestrated **Socket.IO namespaces** to manage multiple concurrent consultation rooms, ensuring isolated, stable real-time data flow.

DairyDesk | Dairy Management SaaS Platform | **Flutter, Node.js, PostgreSQL, Prisma, SQLite (Drift)** [GitHub Repository]

- ❖ Built an **offline-first mobile platform** using Flutter with **Drift (SQLite)**, enabling **60–100 daily entries** with zero internet dependency.
- ❖ Engineered a **bidirectional sync engine** with last-write-wins conflict resolution and **exponential backoff** retry logic.
- ❖ Created an **immutable append-only ledger** tracking milk, payments, and fodder across **10+ relational tables** with full auditability.
- ❖ Applied a multi-strategy authentication system supporting **Google Sign-In**, Email OTP, and offline PIN login using **JWT** and **SHA-256** hashing.
- ❖ Engineered the backend as a **RESTful API** on Node.js/Express with Prisma ORM and PostgreSQL, Dockerized with multi-stage builds and deployed via **Cloud Run** for production scalability.

COURSES & CERTIFICATIONS

- DevOps Fundamentals — IBM (Coursera) · July 2025 [certificate]
- DevOps, Agile and Design Thinking — IBM (Coursera) · July 2025 [certificate]
- Structured Query Language — University of Colorado Boulder (Coursera) · Feb 2025 [certificate]
- JavaScript Basic — Geeks for Geeks · April 2025 [certificate]

ACHIEVEMENTS

- **Competitive Programming:** Solved **500+ problems** (LeetCode, GFG, Codeforces); **Max Rating 1700+ (Top 10% Globally)**.
- **Contest Rank:** Secured **Global Rank 944** in Leet Code Biweekly Contest 162 and 848 in Biweekly Contest 159.
- **Community:** Finalist at **FlutterSprint Hackathon 2K26** · **Google Developer Club**; active participant in EU Code Week, contributing to open-source web accessibility tools.
- **Soft Skills:** Strong communicator with experience in Agile environments and cross-functional team collaboration.